Cassirer's Philosophy of Chemistry

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Introduction

On September 30, 1909, during a visit to Paris, Hermann Cohen, the founder of the so-called Marburg school of neo-Kantianism, sent to Émile Meyerson a postcard asking whether he would be available to meet the following afternoon. A trained chemist, Meyerson (1908) had just published his first philosophical book, *Identité et réalité*, which brought him almost overnight prominence in Parisian intellectual circles at nearly fifty years of age (Bensaude-Vincent and Telkes-Klein 2016). In the same postcard, Cohen noted that one of his former students, Ernst Cassirer, was also in Paris and would join them for the meeting. The following day, October 1, 1909, at around 2 p.m., they met at the Café de la Régence. A few weeks after the meeting, Meyerson received a copy of *Das Erkenntnisproblem*, Cassirer; Cassirer's (1906; 1907) (Meyerson to Cassirer, Nov. 2, 1909; ECN, Vol. 18 (DVD), Doc. 121).[1] Interwoven with the warp of historical detail, the book traces the weft of a quasi-teleological development of modern science from *Substanzbegriff* to *Funktionsbegriff*, an opposition that Cassirer had already begun to articulate systematically in his *Habilitationsvortrag* (Cassirer 1906c).

By late December 1909, Meyerson had already completed the 'first reading' of the *Erkenntnisproblem*, and he had contacted Xavier Léon, the editor of the celebrated *Revue de métaphysique et de morale*, with the intention of drafting a *revue critique* Cassirer's work (Meyerson to Cassirer, Dec. 13, 1909; ECN, Vol. 18 (DVD), Doc. 132). From the outset, Cassirer welcomed Meyerson's proposal with great enthusiasm (Meyerson to Cassirer, Dec. 23, 1909; Vol. 18 (DVD), Doc. 132). Cassirer did not want to pressure Meyerson but he was clearly eager to hear his reaction before and joe that review was, on the second edition of the *Das Erkenntnisproblem*. Meyerson wrote to Cassirer a few months later, in April 19010, after completing the 'second reading' of the *Das Erkenntnisproblem*, announcing a visit to Berlin. Between April 25, 1909 and May 3, 1909, Cassirer welcomed Meyerson into his home on more than one occasion.

[1] The numbering of the Meyerson–Cassirer letters follows ECN, Vol. 18. However, the letters are not transcribed. Scans of the originals are available online at Hamburger Kulturgut Digital; last consulted on October 31, 2026.

The correspondence does not reveal the content of their conversations. Nevertheless, it seems legitimate to suppose that Meyerson anticipated the criticism that would later appear in Meyerson's published review. In particular, Meyerson might already have objected that *Das Erkenntnisproblem* had completely neglected the role of chemistry in the historical development of the 'problem of knowledge' in early modern science. It was Cassirer's exclusive focus on the history of physics that had led him to overemphasize the role of function-concepts at the expense of the substance-concept.

At that time, Cassirer was already in the final stages of drafting *Substanzbegriff und Funktionsbegriff* (Cassirer 1910), whose completion followed not much later, around July 1910. Cassirer immediately asked his publisher to send at least part of the galley proofs to Meyerson (Meyerson to Cassirer, Jul. 17, 1910; ECN, Vol. 18 (DVD), Doc. 148), who, however, received the entirety of the book much later, when the book was already in press . Meyerson was aware, that without having access to it he might have misunderstood Cassirer's stance in the *Das Erkenntnisproblem*. Indeed, he would have definitively realized that, far from neglecting chemistry, Cassirer had chosen to dedicate a rather the 20-page section VIII of the fourth chapter to the topic of concept formation in the natural sciences (Cassirer 1910, **/**).[2] Cassirer's interest for chemistry is far from obvious. Chemistry is conspicuously absent from*Die logischen Grundlagen der exakten Wissenschaften,* the major work of the Marburg school that Paul Natorp concluded at about the same time in March 1910. For Cassirer, by contrast, chemistry has a peculiar epistemological status with respect to physics among the exact sciences. In both cases, scientific formation proceeds from substance-concepts to function-concepts; however, while physics foregrounds the movement toward increasingly general laws, only chemistry brings into view the complementary movement toward increasingly determinate individuality (Cassirer 1910, 270–272; 289–293).

Meyerson's review finally appeared in the January issue of the 1911 volume of *Revue.* Cassirer read it immediately and conceded that from a *historiographical* perspective, *Das Erkenntnisproblem* had given less space to the history of chemistry than was necessary ; indeed, he immediately added an entire new section dedicated to chemistry addressing Meyerson's complaint to the second edition of the *Das Erkenntnisproblem* which appear shortly thereafter (Cassirer 1911). However, from a *systematic* point of view, Cassirer integrated this material into a very different conceptual framework than Meyerson had in mind. What was at stake was ultimately not the relevance of the history of chemistry, but the philosophical conclusions to be drawn from it. On Meyerson's account, the history of chemistry, in contrast to that of physics, unmistakably shows that scientific explanation proceeds by progressively eliminating distinctions until matter is reduced to an undifferentiated unity . For Cassirer, by contrast, the history of chemistry shows more clearly than that of physics that 'generality' required of scientific concept-formation is not an end in itself but serves merely as a means of attaining increasing individual determinacy. Indeed the entire first part of *Substanzbegriff und Funktionsbegriff* Cassirer pointed out concluded the It is chemistry that challenges Heinrich Rickert's (1902) claim that individuality represent the 'limit of scientific concept-formation.'

Reflections on chemistry, from Duhem to Meyerson, played a central role in early French philosophy of science (Bensaude-Vincent 2009). By contrast, German philosophy, perhaps following Kant's own denial of chemistry's status as a 'science,' had always

[2]The number following the slash indicates the corresponding page in the English translation given alongside the original edition in the bibliography.

remained physics-oriented. Cassirer's interest of chemistry represents a unique exception. However, despite the increasing attention given to his philosophy of mathematics and physics, it has received no attention. This paper aims to fill a gap in the literature by reconstructing that background. While Meyerson left an important legacy in France to Metzger and Bachelard, Cassirer's reflections on chemistry seem to have remained isolated in the German context, where even logical empiricism paid no attention to chemistry.[3]

1 Chemistry's Concept-Formation

Cassirer referred to chemistry throughout (Cassirer 1910, ,/b)ut devoted section VIII of the fourth chapter specifically to chemical concept formation (*Begriffsbildung)*). Chemistry had initially developed as a purely 'descriptive science,' much like zoology or botany. However, with the emergence of physical chemistry in the late nineteenth century, it had firmly established itself among the 'constructive sciences'.[4] In particular chemists increasingly turned to thermodynamics to investigate chemical processes, analyzing reaction rates and equilibrium conditions in terms of macroscopic variables such as temperature, pressure, and energy. As Cassirer points out, from this point of view, chemistry's "fundamental laws, as for example the law of phases (*Phasenregel*) of Gibbs or the law of chemical masses (*Massenwirkung*), belong to the same purely mathematical type as the propositions of theoretical physics" (271/204).[5] The difference between physics and chemistry thus lies not in their status as 'exact sciences,' but in the historical paths by which they attained that status. Whereas physics had been a 'constructive' discipline from the outset, chemistry acquired this character only at a later stage in its development.

Physics had the advantage of dealing with relatively simple systems, such as planetary motion, in which bodies could be treated as 'structureless' mass points. It could therefore replace the thing-concept (*Dingbegriff*) with the concept of law (*Gesetzesbegriff*) at an early stage. Chemistry, by contrast, had to begin with the problem of the individual thing (*Einzelding*) in all its complexity (271/204)—specific properties if properties of elements and compounds. Cassirer therefore asks whether chemistry requires a different form of concept formation from that of physics. As one might expect, Cassirer argues that this question can be answered only "if we follow the concrete historical development of chemical doctrines themselves" (272/204). Given the systematic aims of the book, Cassirer does not attempt to trace the history of chemistry in detail, but seeks instead to identify "the great logical lines of direction" along which it developed (272/204). Cassirer does not appear to draw directly on the original sources, relying instead on Wurtz (1879), Meyer (1884), Ladenburg (1902), Duhem (1902) and

[3]Interestingly, Schlick (1932, 108) dismissed Meyerson's philosophy as the philosophy of chemists: "the psychological explanation of [Meyerson's] view lies in the fact that he started as a chemist, who is used to thinking in terms of identical substances".

[4]In support of this point, he could invoke Walther Nernst (1896), one of the founding fathers of modern physical chemistry, who had explicitly articulated this view.

[5]While the 'phase rule' concerns the coexistence of phases and the 'law of mass action' specifies the equilibrium relations among reacting species, both receive a unified formulation within the thermodynamics of Gibbs, Helmholtz, Duhem, and Planck through the notion of thermodynamic potential. Alongside this program of determining the conditions of chemical equilibrium, molecular and ionic accounts of chemical phenomena were already being developed, notably by van 't Hoff and Arrhenius, without yet relying on a theory of the internal electronic structure of atoms (Barkan 1999).

Ostwald (1906) as his principal historical accounts of the development of chemistry. He also draws on Berthelot (1885) in his discussion of the transition from alchemy to qualitative chemistry. Cassirer reorganizes this material to fit his own philosophical narrative. Like the history of physics, the history of chemistry can be understood as a process of progressive 'arithmetization,' in which qualitative differences are expressed in terms of numerical values and the functional relations between them. In chemistry, however, this process serves a different purpose: not the discovery of general laws, but the identification of individual substances.

1.1 From Richter to Dalton

Initially, concept-formation in chemistry is dominated by the classical procces of abstraction of Aristotelian logic: "The properties common to a class of things, which thus provide the basis for forming a particular *genus-concept*, are at the same time singled out as physical constituents and endowed with independent existence" (Cassirer 1910, **/**). Solid bodies differ from liquids and volatile substances by virtue of a specific, separable quality: a change in their state of aggregation entails the disappearance of one quality and the emergence of an entirely different material nature. Mercury, for instance, could supposedly be converted into gold by removing the two 'elements' responsible for its liquidity and volatility and substituting other qualities in their place. In traditional chemistry, elements were conceived not merely as possessing characteristic 'qualities,' but as their very embodiments. Sulfur represented combustibility, salt solubility, and mercury the characteristic properties of metals. Chemical reactions governed by regularities were thus explained by attributing the corresponding qualities to substantial entities. According to Cassirer, this conception of nature persisted into the modern period, most notably in Bacon's physics.

In Cassirer's reconstruction, concept formation in early chemistry followed the traditional model of abstraction, moving from the genus-concept to an ontological conception of substance. Chemists proceeded by reducing complex objects of sensory experience to a set of simple, abstract qualities in terms of which their properties could be explained (**/**). Heat, for example, was regarded as a distinct entity whose presence produced the characteristic effects of warm bodies. Similarly, with the introduction of phlogiston, combustibility, a property that we seem to perceive directly in various bodies, was conceived as a distinct substance admixed with those bodies: "[A]nd from this assumption, the whole structure of chemistry before Lavoisier follows with inner necessity" (205/155). In a footnote to this passage, Cassirer refers to the 'excellent presentation' in Meyerson's (1908) book, which assigns Lavoisier a pivotal role in the quantitative turn in the history of chemistry. Cassirer may well have added this reference at a late stage of revision. Indeed, his account immediately diverges from Meyerson's, assigning a central role instead to the German chemist Jeremias Benjamin Richter. Meyerson, by contrast, does not even mention Richter, although the latter figures prominently in Cassirer's principal historical sources (Duhem 1902, chap. III; Ladenburg 1902, lect. 4; Wurtz 1879, I.II; Ostwald 1906, **).

In Cassirer's view, it was not Lavoisier's law of 'conservation of mass,' but Richter's 'law of definite or constant proportions' that constituted "the starting-point of modern chemical *theory*" (Cassirer 1910, 273/205) and marked a decisive break with the alchemical tradition. Richter's investigations into chemical neutralization helped lay the foundations of quantitative chemistry by establishing, at least implicitly, two principles: that substances combine in fixed quantitative ratios (definite proportions), rather than

in the variable proportions characteristic of physical mixtures, and that these ratios are systematically related across different chemical combinations (reciprocal proportions)[6]. According to Cassirer, as in physics, "[t]he demand for strict numerical determination here also forms the decisive turning point" (Cassirer 1910, 272/205). What matters for Cassirer is not that the permanence of substance takes the form of the conservation (*Beharrung*) of a quantity as in Lavoisier, but rather the establishment of a coordination (*Zuordnung*) between two distinct ordered sequence of numbers.

At this stage, one can establish these laws by measuring the masses of substances involved in chemical reactions "entirely independently of any conception of the constitution of matter, and in particular independently of the atomic hypothesis" (273/205). Following Duhem (1902, 71), Cassirer considers a sequence of acids and a sequence of bases

$$A_1, A_2, A_3, \ldots \qquad B_1, B_2, B_3, \ldots .$$

Each of these substances can be put into correspondence with one of the numbers in the sequence

$$m_1, m_2, m_3, \ldots \qquad n_1, n_2, n_3, \ldots ,$$

so that every acid A_i is assigned a characteristic number m_i, and every base B_j another characteristic number n_j. Suppose sulfuric acid is assigned the number $m_1 = 100$. If 100 grams of sulfuric acid are neutralized by 82 grams of sodium hydroxide, sodium hydroxide is assigned $n_1 = 82$. If 82 grams of sodium hydroxide are in turn neutralized by 74 grams of hydrochloric acid, hydrochloric acid is assigned $m_2 = 74$. Each substance is thus characterized by a number expressing its quantitative combining relations. Once m_p and n_q are known, the relation for the pair A_p, B_q follows: "The manner, in which an element of the first sequence combines with an element of the second, is univocally [*eindeutig*] determined by these numbers; the two weights, according to which an acid A_p combines with any base B_q, are related according to the corresponding numerical values m_p and n_q" (Cassirer 1910, 273/205).

Richter tabulated the quantities of the bases which are neutralized by the same weight of sulphuric acid, of hydrofluoric acid, etc., and these he calls neutrality sequence, or sequence of masses; he also determined the quantities of the acids which are saturated by. the same quantity of different bases. In doing this, he thought he had discovered certain regularities. The sequence of masses in the case of the bases formed an arithmetical, and that in the case of the acids, an arithmetic progression. This particular hypothesis turned out to be empirically false (Ladenburg 1902, **; Wurtz 1879, **; Ostwald 1906, **). Like Duhem (1902, 144), however, Cassirer aims to emphasize that Richter borrowed nothing from atomistic doctrines; his numerical speculations instead expressed the same "general Pythagorean doctrine of the 'harmony' of the cosmos" (Cassirer 1910, 273/206) found in Kepler's work. In Cassirer's characteristic rhetoric, this case illustrates how scientific concept-formation does not primarily consist in discovering what each of these things 'really is' internally. It consists instead in finding a rule that assigns each member a determinate place within an ordered system.

As the conventional rational recontruction goes, Richter's scheme ran into a difficulty: the same two elements can form more than one compound, each with its own definite

[6]The attribution varies across historical accounts. Ostwald, for example, attributes the law of definite proportions to Proust and the law of reciprocal proportions to Richter.

composition. In other terms, they do not couple in one unique set of proportions. Carbon and oxygen, for example, form both carbon monoxide and carbon dioxide. In the former, 12g of carbon combine with 16g of oxygen; in the latter, 12g of carbon combine with 32g of oxygen. A single proportional number for oxygen cannot account for this variation. It was John Dalton to provide a solution to the problem formulating the 'law of multiple proportions.' Instead of discarding the idea of characteristic masses (p_i), he retained them as invariant base units and introduced integer multipliers (λ, μ, ν). The mass ratio of any compound is then no longer simply $p_1 : p_2$, but

$$\lambda p_1 : \mu p_2 : \nu p_3 : \cdots$$

where p_i is the invariant characteristic mass of the element (e.g., $p_C = 12$, $p_O = 16$) and λ, μ, ν are whole numbers $(1, 2, 3, \ldots)$. In carbon monoxide, the mass ratio is $1 \cdot 12 : 1 \cdot 16$, so that $\lambda = 1$ and $\mu = 1$; in carbon dioxide, it is $1 \cdot 12 : 2 \cdot 16$, so that $\lambda = 1$ and $\mu = 2$. This is the origin of the Berzelius's *compositional* formulas, which express the proportions in which elements combine to form compounds. On this purely quantitative reading, a formula such as

$$\mathrm{H_2O} \quad \text{or} \quad \mathrm{CO_2}$$

is an algebraic shorthand for these integer multipliers. The letters represent the characteristic proportional masses (p_1, p_2), and the subscripts the integers (λ, μ). Formulas of this kind allow not only to summarize given empirical experiences, to elicit new ones: Arranging their formulas into a numerical sequence might expose a gap: oxygen occurs in quantities of 1, 3, and 4, suggesting a compound with 2.

The goal "law of constant combining numbers" (*Verbindungszahlen*) was initially to characteristic equivalent number for each element (*Äquivalenzzahl*) and that whenever two or more elements enter into a compound, their masses stand to one another "as whole multiples of these basic numbers" (Cassirer 1910, **/**). In principle, one can establish the laws of definite, reciprocal and multiple by measuring the macroscopic masses of substances involved in chemical reactions. In Dalton's work, however, this rule of 'multiple proportions' immediately merges with a particular interpretation: "The concept of combination-weight [*Verbindungsgewichts*] is transformed into that of atomic weight [*Atomgewichts*]" (274/206). If matter were continuous, like a fluid, elements should be able to combine in any proportion whatsoever. However, according to the law of multiple proportions, 12 g of carbon combine with exactly 16 g or exactly 32 g of oxygen. The multipliers (λ, μ, ν) are whole numbers; there is no intermediate possibility. Dalton reasoned that if substances always combine in fixed mass proportions, it must be because they consist of indivisible units, namely atoms, with precisely those relative atomic weights. Different elements possess atoms of different masses, while all atoms of the same element possess the same invariant mass: "Statements concerning the essential nature of the ultimate constituent parts of individual bodies have replaced the empirically determined numerical proportions (*Proportionalzahlen*) of those bodies" (**/**) Dalton suggest that chemical mass proportions could be used to calculate relative atomic weights by taking Hydrogen = 1 as the unit standard.

Also, in the next stage, Cassirer follows the conventional account. Dalton's hypothesis in turn faced a fundamental difficulty. The empirical facts do not determine the relative atomic weights uniquely. Suppose hydrogen is taken as the unit: the same experimental composition can be represented using different atomic weights for oxygen, provided

that the molecular formulas are altered accordingly. Thus the experimental facts of composition alone cannot decide between, for example, $O = 16$ and $O = 8$. The same applies to sulfur, for which values such as $S = 8, \quad S = 16, \quad S = 32, \ldots$ are equally admissible (Cassirer 1910, 274–75/206–7). Chemists could have accepted this ambiguity as a fact; instead, they introduced additional, independent relations to determine the ratios unambiguously. Following Meyer's (1884) account, Cassirer points out how the *univocal* (*Eindeutig*) the relative atomic weights is progressively determined not by a single measurement, but by the mutual convergence and correction of independent physical laws. Cassirer, following Meyer, mentions three especially important ones: Avogadro's rule,[7] connected with gas volumes and vapor densities; the Dulong–Petit law, connecting atomic weights with heat capacities[8]; and Mitscherlich's law of isomorphism, connecting crystalline form with chemical composition and structure[9] (Cassirer 1910, 275/207). No single rule can univocally determine the relative atomic weights; their strength lies in their capacity to correct one another's anomalies. Relative atomic weights become increasingly determinate because several initially distinct sequence of phenomena are made mutually consistent: "It is only the totality of all these different points of view, mutually confirming and correcting each other, that finally after many experiments gives a unitary table of atomic weights, and thus lays the basis of a definite system of chemical formulae" (275/207).

For Cassirer it becomes clear how the "the conditions of scientific *production* differ from those of critical *reflection*" (**/**). A modern chemist might maintain that "[t]he objective existence of the different types of atoms is presupposed; it is only necessary to discover their *properties*, and to define them more exactly" (275/207). As inquiry advances and encompasses an increasingly diverse range of phenomena, atomic properties appear to become richer and more determinate, as though the underlying substance of atoms were gradually revealing itself in a concrete and tangible form. Although scientists may regard the existence of atoms as their starting point, Cassirer, following Duhem and Ostwald, argues that a closer examination of the empirical grounds for their claims reveals precisely the opposite. One starts with a series of experimental laws (the law of the conservation of mass, definite, reciprocal and proportions) that reveal that, Unlike physical mixtures, chemical compounds have fixed compositions by mass. At this point, concept of 'combining weight' naturally transforms into the concept of a discrete 'atomic weigh,' as arising from whole-number combinations of discrete units: "It becomes clear that the atom is never the given starting-point, but always only the goal of our scientific statements" (276/208).

Cassirer rejects Wurtz naive atomism about atom, however, he rejects Duhem's anti-atomism. For Cassirer, the alternative is misleading. The question lies not in the "metaphysical claims about the existence of atoms" but in "the logical function

[7]The Principle: Equal volumes of all gases under the same pressure and temperature contain equal numbers of molecules ($M \propto d$). By measuring vapor densities across a wide array of volatile compounds containing a given element, chemists determine the smallest mass quantity of that element present in two volumes of gas. This sets the definitive upper boundary for the element's atomic weight.

[8]The product of an element's solid specific heat and its atomic weight is approximately constant ($A \cdot c \approx 6.0$–6.4): For non-volatile solid elements—especially metals—where vapor densities cannot be measured, Dulong and Petit's law provides a direct thermal determination of atomic weight.

[9]Compounds possessing identical crystalline forms contain the same number of atoms combined in the same structural Application: Where neither vapor density nor specific heat is accessible, an unknown atomic weight can be deduced by substituting the element into an isomorphic matrix of a known compound (e.g., perchlorates and permanganates).

that the concept of the atom [...] performs" (Cassirer 1910, **/**). If one confines oneself strictly to observational data, macroscopic experience one might accept as a matter of fact that elements combine in several distinct ratios. Daltonian atomism, by contrast, forced chemistry to resolve the ambiguity by resorting to a combination of independent physical laws.[10] While we appear to investigate the atom itself and its weight as a property, they instead used the sequence of atomic weights as a mediating series that makes it possible to coordinate otherwise unrelated domains of experience. The Avogadro–Cannizzaro method, for example, establishes a relation between the vapor densities of gaseous substances and their molecular weights, thereby providing an independent criterion for determining atomic weights:

$$\rho_1, \rho_2, \ldots, \rho_n \,,$$

to their molecular weights, relative molecular weight of any gaseous substance is directly proportional to its gas density, from which the atomic weights of their constituent elements,

$$a_1, a_2, \ldots, a_n \,,$$

can be determined by taking chemical composition into account. On the other hand, the Dulong–Petit law establishes a correspondence between these same atomic weights and the specific heats of solid elements,

$$b_1, b_2, \ldots, b_n \,,$$

so that the product atomic weight and spefici weight is a fixed constant. The same series of atomic weights a_n thus connects two apparently unrelated empirical domains, the behavior of gases and the thermal properties of solids (276–77/208).

The question of the reaolty of atoms, this is here replaced by their succes of the atom-concept in organizing in establish a univocal corodatn beteen otherwise unrealtied series: "The atom functions here as the conceived unitary center of a system of coördinates, in which we conceive all assertions concerning the various groups of chemical properties arranged" (277/208). The various and initially heterogeneous numerical sequences acquire a univocal coordination (*eindeutige Zuordnung*) as we refer each of them back to a mediating sequence of atomic weights. The atom with a determinate atomic weight is not the underlying 'bearer' of different properties that we discover; rather, the sequence of atomic weights is the "common sequence of comparison" mediating between different numerical sequences: "In truth, we are concerned not so much with relating the diverse sequence [*Reihen*] to the atom, as rather with *relating them reciprocally* to each other through the mediation of the concept of the atom" (277/208).

The empirical laws of multiple combination motivate the introduction of *discrete* atomic units, while other laws make it possible to determine their relative weights and integrate them into a common system. The attempt to determine unambiguously (*eindeutig zu bestimmen*) the atomic weight of each element requires drawing upon ever new empirical laws governing different domains of chemical and physical phenomena as criteria (**/**). As this determination progresses, the range of empirical relations itself expands accordingly: "If we imagine this progress brought to completion, then the totality of possible relations that the individual series can enter into with one another

[10] Here Wallston's chemical equivalents and Dalton's atomic weights discused (Ladenburg 1902).

would be encompassed and expressed precisely in the 'absolute' atomic weights", or better the absolute value of the relative atomic weights (Cassirer 1910, **/**). On the other hand, by attributing observations concerning vapor density, heat capacity, isomorphism, and so forth to one and the same subject, we organize disparate known empirical laws into a system. On the other hand, in the attempt to determine the atomic weights, we discover new empirical laws that were previously unknown. Cassirer can thus conclude with a characteristic neo-Kantian move: the atom should not be conceived as the 'substantial unity' from which scientific investigation begins, but rather as the 'projected unity' toward which it strives: "Thus, the chemical atom is an 'idea' in the strict sense Kant gave to this term" (**/**).

1.2 From Substitution to Valence

From a logical point of view, one can say that the first stage in the development of chemistry was the ability to determine the relative atomic weight of elements considered individually. In Cassirer reconstruction, the second stage was to establish how they are systematically ordered in relation to one another, arranging them into classes of a distinctive character. Following his sources, Cassirer points out that "[t]he empirical facts that first lead to such relative distinctions and groupings within the overall system are given in the relations of chemical *substitution*" (**/**). It was Jean-Baptiste Dumas who noted that for every volume of hydrogen removed from a compound in acetic acid by chlorine, an equal volume of chlorine is taken up producing trichloroacetic acid which reacted in almost identical ways to the original. This result challenged the Berzelius's dominant dualistic assumption that molecules must be split into positive and negative halves so that replacing a 'positive' hydrogen atom with a 'negative' chlorine atom was physically impossible (**/**).[11] Yet as the theory of substitution gains ground, the opposite view increasingly prevails: even entirely dissimilar substances can replace one another in certain compounds without altering the nature of the compound. Examining how different elements substitute for one another reveals basic rules governing these substitutions: "The form of substitution can be fixed and expressed once and for all by means of certain *numerical values* that we assign to each element and that are now added to the number of its combining weight" (**/**).

To create a mathematical system, one needs a baseline of 1. Just like hydrogen's mass is the baseline for equivalent weights, hydrogen's substitution capacity is the absolute minimum combining capacity. If one wants to insert oxygen into a chemical framework, the cost of admission is the removal of exactly two hydrogen units. This establishes the relational ratios: $Cl \sim 1H, O \sim 2H, N \sim 3H, C \sim 4H$. This establishes a further basis for relating the individual elements to one another, a "new characteristic constant" for each elementary substance belongs to them independently of their chemical affinity with which substances does it tend to combine: "The 'valency' of the elements is the expression of a certain fundamental property in them, which belongs to them independently of their chemical affinity" (**/**). Without the concept of valence, one might merely record individual cases in which atoms of one element replace those of another in compounds. Valence brings them under a general rule, allowing one to determine which substitutions preserve a given pattern of bonding.

The 'general rule' is that substitution must preserve the total valence: If hydrogen's base value is set to 1, oxygen's value is 2. Therefore, if you want to replace oxygen in

[11] On Berzelius see (Ladenburg 1902, lect. VI).

a compound using hydrogen, you must supply twice the volume. Once again it was realized, that chemical substitution does not occur in arbitrary fractions; it operates *per saltum*. One atom of Hydrogen is replaced by exactly 1 atom of Chlorine, or 2 atoms of Hydrogen are replaced by 1 atom of Oxygen and so on. Arranged according to this principle, chemical compounds fall into various 'principal types' (*Haupttypen*) (Cassirer 1910, **/**). Compounds belonging to the same type can be derived from a common basic form through successive substitutions governed by the valences of the atoms or groups involved. Substitution thus becomes an instrument of classification. Methane gas provides a simple illustration. In this bulk substance, carbon exhibits a total combining capacity of four, which is entirely satisfied by a proportional mass of hydrogen. Because chlorine shares the exact same exchange rate (valence) as hydrogen, one can progressively react methane with chlorine to generate a series of new substances. Each successive substitution strictly preserves carbon's overall fourfold requirement:

$$\mathrm{CH_4} \longrightarrow \mathrm{CH_3Cl} \longrightarrow \mathrm{CH_2Cl_2} \longrightarrow \mathrm{CHCl_3} \longrightarrow \mathrm{CCl_4}.$$

Despite exhibiting drastically different physical properties in a flask, because each step involves trading exactly one equivalent of hydrogen for one equivalent of chlorine, these compounds are chemically understood as proportional modifications of the same fundamental 'methane type.' The same classificatory principle is illustrated by the water type. In the water type, H−O−H, divalent oxygen provides two bonding positions. Replacing one hydrogen atom with an appropriate monovalent organic group yields an alcohol, R−O−H; replacing both yields an ether, R−O−R′. Despite their different properties, water, alcohols, and ethers share the same basic pattern of two attachments to oxygen.

Following the logical rather than historical order, Cassirer could then treat the notion of valence as the fundament of the notion of 'chemical type.' What interests Cassirer is the fact that 'substitution' represents the exact moment chemistry abandoned traditional 'classification' generic category-concepts (*Gattungsbegriffe*) and mimicked mathematical 'classification' based on function-concepts (*Funktionsbegriffe*):

> Here again, the concept of 'type' comes into consideration not in its significance for the special problems of chemistry, but only as a paradigm for certain logical relations. Indeed, it exhibits with full clarity the characteristic fundamental features that the analysis of the exact scientific concept has already established in general. The chemical concept of type, too, is formed not on the model of generic concepts but on that of serial concepts. The various compounds subsumed under a principal type owe this grouping neither to an external similarity in their sensible properties nor to an immediate kinship in their chemical functions. They belong together insofar as they can be transformed into one another by virtue of the relation that holds between the valences of the individual atoms. The more remote members of the series, however, need not offer the nearer ones any *analogy* other than the one established by this very law of derivation. (**/**)

Dumas still grouped compounds derived from one another by substitution that retained similar main chemical properties; Regnault extended the notion of substitution to encompass a whole sequence of reactions resulting in products that possessed completely different properties from the initial compound. The unity of the chemical type rests not on the similarity or analogy of the substances themselves, but on the rules governing their possible transformations through substitution governed by 'valence.' Here the *conceptual abstraction*, as conceived by traditional logic, establishes the unity of a class by identifying a characteristic common to its members, *mathematical abstraction*

establishes conceptual unity through their mutual substitutability *salva veritate*. In geometry, two figures, however different they may appear at first sight, are regarded as equivalent if one can be obtained from the other through transformations belonging to the same group. Similarly, in chemistry, two compounds may be regarded as belonging to the same chemical type if one can be derived from the other, directly or through a succession of substitutions, even when their chemical functions and properties differ considerably.

Once again inverting the historical order, Cassirer concludes that the implications of this result become particularly evident when we turn to organic chemistry, as the theory of substitution acquired greater scope when chemists recognized that groups of atoms (*Atomgruppen*), as well as individual atoms, could function as substitutive units. The conversion of methanol into chloromethane,

$$CH_3OH + HCl \longrightarrow CH_3Cl + H_2O\,,$$

illustrates the persistence of a compound group through a chemical transformation. Here, CH_3- functions as a compound radical (*zusammengesetztes Radikal*) as Liebig defined it, while OH is replaced by Cl. A radical could replace one or more hydrogen atoms in a compound and was accordingly described as monovalent, divalent, and so on. For example, methyl (CH_3-) has one point of attachment, whereas methylene (CH_2-) has two, and so on.[12] Whereas Laurent considered radicals as 'nuclei' possessing an independent existence in the compound, Gerhardt treated them as 'residues' emphasized that their logical function of remain constant in a series of compounds (Cassirer 1910, **/**). In the expression CH_3-, CH_3 represents the radical, treated as constant across a series, while the empty place $-$ represents a replaceable atom or group. The empirical reality of 'radicals' is not they correspond to *actual* thing-like entities, but resolves into their logical function: they identify what remains invariant in a compound through possible substitutions, thereby allowing us to anticipate the possible behavior if substitution were performed suggests further possible members of the same series: "The whole that we seek and toward which the concept is directed must not be conceived as an absolute being outside all possible experience; it is nothing other than the ordered totality of these possible experiences themselves" (**/**).

1.3 The Periodic System

Establishing a consistent scale of relative atomic weights was itself chemistry's substantial achievement. As Cassirer put it, each individual element thereby acquired, in Leibniz's expression, a 'characteristic number,' implicitly regarded as expressing its distinctive empirical character (287/**). Whereas atomic weight distinguished the individual members, valence, an integer count of saturation units, provided a principle for classifying them according to their combining capacities, as distinguished from the affinity force of combination, measured by heat release. One might be tempted to explain these empirical facts by treating substances as made up of atoms as solid spheres of determinate weight endowed with physical 'hooks' —if carbon had four hooks, they assumed it must always attach to four things. However, a mechanical, erector-set model worked for early organic chemistry because carbon is famously consistent in how it bonds, but as inorganic chemistry advanced, it revealed that elements like

[12] Although these concepts were historically discovered in the reverse order, their logical dependency unfolds in precisely the sequence identified here, culminating in substitution.

nitrogen, phosphorus, and sulfur have variable valence. The real advance of notions like atomic weight, valence etc. is that the properties of as propertys of atoms as thinksl, into positons in the numerical series mediate the conversion of a qualitative concept into a position in a numerical series in which each element is defined by a pot "This requirement now extends to all physical and chemical determinations recognized as dependent on definite numerical values [*Zahlwerten*]" (Cassirer 1910, **/**).

Here the "methodological advantage of the domain of numbers" (*Zahlgebiet*) becomes apparent (**/**). Chemistry undergoes the same process of arithmetization that characterizes the history of physics, a process that culminates in the establishment of the *periodic system*. Drawing on the more consistent atomic weights made available by Cannizzaro's 1858 reform, Cassirer attributed to Lothar Meyer the merit to have published his systematic table arranged the elements by increasing atomic weight and grouped their valence (combining capacity). In the 1884 edition of *Die modernen Theorien der Chemie*, that Cassirer is referring to, Meyer arranged the elements in rows of increasing atomic weight from left to right, then aligned the rows so that elements with similar chemical properties fell into vertical columns, each representing a family of elements with the same *highest possible* combining capacity or valence. By arranging elements of equal valence in order of increasing atomic weight, Meyer revealed that the atomic weight differences (ΔA) between analogous elements in a family follow step-like numerical progressions: approximately 16 for light elements, 46 for medium elements, and 88–92 for heavy elements. This table showed that chemical properties, specifically how elements bond with hydrogen or oxygen, repeat at regular 'periodic' intervals.

Meyer plotted the physical atomic volume V of the elements (calculated by dividing atomic weight by density in the solid state) against their atomic weight (A). Connecting these points produced a dramatic, wave-like periodic curve $V = f(A)$. Elements that occupy analogous positions on this curve, whether on its ascending or descending branch or at one of its maxima or minima, also prove to be analogous in their most important chemical and physical properties (valence, volatility, ductility, electrical and thermal conductivity, and so on). The high peaks of the curve were occupied by the light, low-density alkali metals (lithium, sodium, potassium, rubidium, cesium). The low troughs were occupied by dense, heavy transition metals (such as iron, cobalt, nickel, platinum, and osmium. Other physical properties—such as ductility, fusibility, volatility, thermal expansion, and electrical conductivity—increase and decrease in exact synchronization with this volume wave. As Cassirer emphasizes, Meyer (1884, 176) himself observes that 'matter' has now passed from the domain of the constants of natural science into the domain of the variables. Instead of regarding the characteristic properties of each element as independent *constants*, chemists could now treat them as functions of a common *variable*. This ordering relation is scientifically meaningful independently of whether atoms are understood as physically existing entities: "The qualitative nature of the particular material is made mathematically conceivable by discovering a point of view from which the material can be arranged in sequence with a definite law of progression" (Cassirer 1910, 289/217).

Chemistry is now able to identify individual chemical elements as discrete arithmetical entities defined by their position in the periodic system. At this stage, the relative atomic weight was assumed to be the ultimate, immutable, discrete constant (*starre Einzelwerte*). If atomic weights remain static, discrete constants, science is forced to accept them as unexplained empirical 'givens.' Yet the constancy of relative atomic weights cannot be proved to be ultimate. Indeed, Cassirer could point to the discovery

of phenomena like radioactivity, which seems to suggest that the chemical atom might have its status as an ultimate, indivisible entity and "dissolves into a system of electrons" (Cassirer 1910, **/**). All measurement methods led, with increasing accuracy, to the same values of the charge and mass of an electron. In this convergence, science might have found a new "relative point of stability" (**/**). However, the new constant thus obtained might turn out to be a variable. (**/**). Cassirer can therefore conclude that scientific inquiry never establish 'atoms' as absolute elements of reality independent of any theory; the empirical reality of the 'atom-concept' is the ideal limit of the series of changing theories about the atom: "The convergence of the series (*Reihe*) takes the place of the external criterion of reality" (**/**).

2 The Polemic with Rickert. The Case of Isomerism

Just as physics replaces the classification of perceptible colors with a numerical ordering of monochromatic light according to wavelength, chemistry replaces the classification of substances by their qualitative properties with a numerical ordering of elements according to their relative atomic weights. However, Cassirer insists the chapter on chemistry was not simply as an appendix to previous sections on physics. Chemistry reveals an aspect of scientific concept formation that an exclusive focus on physics would have left unexplored. As Cassirer pointed more opening the next section, "[t]he real *methodological* interest of the construction of the concepts of chemistry lies in the fact that in it the *relation of the universal to the particular* is set in a new light" (Cassirer 1910, 292/221). When we consider physics in isolation it seems that to generate the traditional antinomy: 'knowledge' aims at the universal in the form of laws, while genuine 'being' belongs to individual substances. Individual phenomena enter the picture chiefly as instances through which those laws are illustrated. From this perspective, one might reasonably conclude that increasing scientific generality entails a progressive neglect of individuality. According to Cassirer, in contrast to physics, chemistry seems from the beginning to try to grasp the qualitative features of individual things (*Einzelding*) (such as gold, water, or specific organic compounds).

Cassirer's aim in his discussion of chemistry was to show that these two directions of concept formation in the natural sciences are mutually compatible and, indeed, inseparable. It is worth noting that the expression 'concept-formation' (*Begriffbildung*) does not play any role in the other works of Marburg school. For example, the term *Begriffsbildung* does not compare at all in Natorp 1910 book. On the contrary, prominence is given to this term by the so-called Heidelberg-Freiburg school. In particular Heinrich Rickert (1902)[13] as famously to emphasis 'limits of concept formation.' The more universal a scientific law or concept becomes, the further it moves from actual reality, creating an unbridgeable gulf between natural scientific concepts and actual empirical reality. To grasp reality in its uniqueness, human knowledge must adopt a different, complementary methodology—the individualizing method of the historical and cultural sciences. By framing his work in terms of the debate over scientific concept formation, Cassirer clearly sought to expose the inadequenci of Rickert's sharp distinction between the *generality* of natural science and the *individuality* of history. However, it is only after treating chemistry that Cassirer was in the position to launch a direct attack against Rickert in the concluding section IX of the first part of the book.

If Rickert's reasoning is sound, Cassirer continues, scientific research has hitherto

[13]Windelband in Heidelberg and Rickert in Freiburg.

labored under a peculiar misconception about its own ultimate goal. Science seeks to know 'reality,' yet the very instrument it employs, the natural-scientific concept, appears to frustrate this aim: "What was intended to sharpen our grasp of the particulars of empirical intuition must now be recognized as blunting it. To grasp reality conceptually is thus to destroy its most distinctive and fundamental content" (Cassirer 1910, **/**). According to Cassirer, however, this conclusion follows not from the nature of scientific concept-formation as such, but from the premises underlying Rickert's account of the latter. As one might expect, according to Cassirer, Rickert remains implicitly committed to the prevailing view of the universal concept is ultimately genus-concepts *Gattungsbegriff* a representation of what is common and therefore as incapable of grasping the particular as particular. Following Sigwart (1889) Rickert claims that concept-formation is nothing but an extension of word-formation: Just as, for instance, no fully determinate intuitive content corresponds to the word 'bird,' but rather only certain vague outlines of shape are retained in it together with the indistinct representation of the movement of flight, so that a child may also call a flying beetle or butterfly a bird |, so the same holds originally for all our general representations.. General scientific concepts also arise from indeterminate, faded representations that preserve only a few shared characteristics.Scientific concepts make classification more precise, but on this model they merely refine the same abstractive procedure, moving still further away from concrete intuition.

For Cassirer, however, exact scientific concept formation is not model on the refining the meanings of words, but it follows the logic of modern mathematics. The fundamental logical model of scientific thought is the sequence-concept (*Reihenbegriff*), whose prototype is the number sequence. Within the sequence of natural numbers, each number is a *distinct* 'individual,' uniquely determined by its position in the sequence. Yet all natural numbers are generated through the repeated application of one and the same rule, the successor function. A determinate number thus breaks with the traditional logical schema, which recognizes concepts only as general genus concepts subsuming a plurality of instances. 'Two' and 'four' do not exist as genera instantiated in all concrete pairs or groups of four objects of the same succession function. For Cassirer, numbers provide the paradigm of conceptual individuality. The transition from the individual to the general concept does not involve disregarding particular differences. Generality is thus achieved not through the progressive omission of differences, but by transformation of a constant into a variable. The individual thus appears not as something left behind by the general concept, but as a particular determination within it.[14]

The same logical principle underlies the formation of general physics concepts. Consider a particular quantity of gas. Initially, we might describe it in terms of its indefinitely many perceivable characteristics. Physics first replaces the real gas with an idealized model whose behavior is characterized by only a few variables: pressure p, volume v, and absolute temperature T. A possible state of the gas is represented by a triple (p, v, T), whose components range over appropriate domains of possible numerical values ($| \; p > 0, V > 0, T > 0\}$). The here mathematically possible does not yet mean physically admissible (Cassirer 1910, **/**). The choice of these state variables reflects the fact that their values are connected by a functional relation, expressed in the ideal

[14]Cassirer can draws on Rickert's student Lask (1902) acknowledges that mathematical concepts possess a 'concrete universality': unlike ordinary general concepts, they can generate their individual instances through construction.

gas law as

$$pv = RT\,,$$

which restricts those candidate states. The ideal gas is thus an abstract representation of a real physical system: it describes how a gas *would* behave under idealized conditions in which additional physical effects are neglected. In this model, all gases are treated as equivalent. Their substance-specific characteristics are disregarded, and their behavior is represented by the same functional relationship. At this point, Rickert's objection might still appear justified: "scientific research cannot do without, alongside | the 'given' elements of perception, purely ideal limiting concepts that cannot be exhibited in any direct experience; that it cannot do without the 'unreal' alongside the 'real.' ". (Cassirer 1910, **/**)

Rickert might therefore argue that the formulation of universal laws requires the elimination of individual differences. According to Cassirer, however, scientific concept-formation proceeds in a very opposite direction, it does not remove individual differences but it transforms them into numerical variables. To account for the differences between 'real' gases and the 'ideal' gas, physics moves to a more comprehensive functional relation that encompasses the ideal gas law as a special case. For example, the van der Waals equation,

$$\left(p + \frac{a}{v^2}\right)(v - b) = RT\,,$$

introduces two additional empirically determined constants, a and b, whose values differ from one gaseous substance to another.[15] These constants represent, approximately, intermolecular attractions and the finite volume occupied by molecules, allowing the same mathematical framework to distinguish substances whose behavior the simpler model treated as equivalent. The ideal gas law is recovered by setting $a = b = 0$. Thus, the introduction of additional parameters does not merely refine the description of a particular gas; it transforms the original law into a special case of a more comprehensive family of possible functional relations. The more general mathematical framework thereby permits a more differentiated empirical characterization of particular gaseous substances: "Auch von dieser Seite her zeigt es sich somit, daß dem wissenschaftlichen Begriff als solchem | die Feststellung des Einzelnen keineswegs versagt ist" (**/**). The movement toward greater systematic generality is simultaneously a movement toward the more precise determination of the particular.

The example of the gas laws for the primary task of concept formation is not increasing indeterminacy but rather bringing it to increasing determinacy. Rickert might still object that even the 'real gas' thus conceived is not a genuinely concrete 'individual.' According to Cassirer, however, this objection loses much of its force once we move from physics *laws* of physics to the *formulas* of chemistry. To establish this point, Cassirer appeals to the phenomenon of 'isomerism' . Before the recognition of isomerism, chemists generally assumed that when two distinct substances share the exact same empirical or *compositional* formula

$$\mathrm{C_2H_6O}\,,$$

[15] I borrow this example from a later writing (Cassirer 1936a, **; tr. 1956, **).

they are 'indistinguishable' at the level of elementary percentage. Yet cases were discovered in which allegedly compositionally identical substances, in analogous circumstances, give rise to different reactions, undergoing different substitutions. If chemistry followed the concept-formation traditional logic, trying to grasp the individual difference between the two isomers would require adding more *concrete* sensory descriptions (color, boiling point, odor). On the contrary, chemistry proceeds in the opposite direction, introduced more *abstract*, *constitutional*, or structural formulas, which takes into account "how atoms are arranged alongside one another and mutually linked to form the unified structure of the molecule" (Cassirer 1910, **/**):

$$CH_3 - O - CH_3 \qquad CH_3 - CH_2 - OH$$

specifies how those atoms are bound together through valency units to form the molecular framework. Thus, individual differences of isomers are grasped by introducing a moving beyond simple weight to include the spatial configuration of atoms:

> Thus, for example, the chemical "concept" of a particular substance is given by its structural formula, in which it is conceived as a specific substance with its distinctive structure, while at the same time being assigned to the various chemical "types" and thereby placed in a definite relation to the totality of other substances. The ordinary chemical formula, which indicates only its general composition but not the manner in which the individual elements are arranged, is here enriched by a wealth of new relations. The general rule now at our disposal also allows us to trace how and according to what law the given substance is transformed into others: it encompasses not merely the form of its existence at a particular moment, but the totality of its possible spatial and temporal phases. The further chemical concept formation advances, the more sharply its capacity to distinguish particular substances becomes articulated. Substances that, from the standpoint of the less developed concept, had to be regarded as alike because they were "isomeric" become clearly differentiated from the standpoint of the more developed concept and are distinctly demarcated in their individual characteristics.. (**/**)

But the processes were not concluded. Compounds were found to share the same structural formula while exhibiting opposite optical behavior.[16] Optical isomerism thus revealed the limitations of two-dimensional structural formulas, just as constitutional isomerism had exposed the limitations of compositional formulas. Once again, the *individual* differences between substances could be captured by introducing more *general* forms of representation. Van 't Hoof replaced the two-dimensional structural formulas with three-dimensional stereochemical formulas. Cassirer resisted the idea that these formulas represent the actual spatial arrangement of atoms, they encode the range of possible behaviors of compounds under different chemical conditions: "the constitutional formula is, precisely thereby, also the genuine scientific expression of the empirical reality of the body: for this reality signifies nothing other than the thoroughgoing objective connection in which an individual 'thing' or a particular event stands with the totality of actual and possible experiences in general" (**/**)

The case of chemistry shows that, *pace* Rickert, conceptual progress in science does not consist in disregarding particular differences in favor of increasingly abstract generalities. On the contrary, the development of more comprehensive relational systems allows for the increasingly precise determination of particulars: "The individual in its peculiarity is threatened only by the universality of the blurred generic image, while the

[16] The so-called enantiomer; one enantiomer rotates plane-polarized light to the right, while the other rotates it by exactly the same angle to the left.

universality of a definite law of relation confirms this peculiarity and makes it known on all sides" (Cassirer 1910, 299/225). Rickert might object that an individual chemical compound is a repeatable chemical kind rather than an individual in the strict sense. However, this objection would amount to conceding that his real concern is not that scientific concepts fail to capture individuality (*Einzelheit*), but that they fail to capture the concrete 'thinginess' (*Dinglichkeit*) of the objects of common-sense experience (**/**). These, however, are distinct problems. What counts as an 'individual' is not given to theory as an external 'thing,' but is determined within the theory itself. Different forms of inquiry—chemistry, biology, history,[17] and even everyday experience—employ different criteria for individuating their objects. In this sense, and only in this sense, the integration of these different forms of 'individuality' can never be completed. As one might expect from Cassirer, it remains an infinite task: "Here, individual reality indeed reveals its fundamental character of inexhaustibility. Yet it is precisely the distinctive virtue of genuine scientific relational concepts that they undertake this task despite its being, in principle, incapable of completion" (**/**).

3 A battle over the Substance-Concept. The Meyerson–Cassirer Debate

In the summer of 1910, Cassirer asked his publisher to send Meyerson the galley proofs of *Substanzbegriff und Funktionsbegriff*. The com proofs of Cassirer's monograph were delivered with considerable delay. As Cassir As once Cassirer they could continue their "battle over the substance-concept" (Cassirer to Meyerson, Aug. 27, 1910; ECN, Vol. 18 (DVD), Doc. 153), was by the editorila and full their recepti pos. When Meyerson received the complete proofs of *Substanzbegriff und Funktionsbegriff*, his review of the *Erkenntnisproblem* was already in the course of publication in the *Revue de métaphysique*. In a letter from October 1910, Meyerson wrote to Cassirer: "I am now in possession of your *Substanzbegriff* in its entirety, albeit without the cover page. This is, however, unessential, since I shall in any case have the book bound. The (very long) article on your *Erkenntnisproblem* has for some time been lying corrected at the editorial office, but, as I was recently informed, it may not yet appear in the next issue, for typographical reasons!" (Meyerson to Cassirer, Oct. 21, 1910; Vol. 18 (DVD), Doc. 157). The publication of Meyerson's review took several more months, and the review did not appear until January 1911.

3.1 Meyerson's Review of the Das Erkenntnisproblem

In nearly thirty pages, Meyerson surveys both the first and second volumes of the *Erkenntnisproblem*, offering several comments and critical remarks on each section. The review begins with a generally positive assessment: "Cassirer's book—as he himself warns us—is only apparently a book of history. In its essence, it is a systematic book, a work dominated by a theory that the author seeks to support through the study of the philosophical and scientific evolution of the modern period. This prevalence of a unified

[17] Cassirer draws on a recent essay on individual causality by Hessen (1909) to argue that even historical causality contains an element of universality. To explain the outbreak of a war, for example, historians routinely invoke a heterogeneous set of psychological, political, economic, institutional, and geopolitical regularities. Yet whereas the natural scientist's primary objective is to establish generally valid relations, the historian's task is to determine which regularities are operative in a particular case and how they interact to produce a unique outcome. The universal and the individual are therefore not opposed even in historical inquiry. On the contrary, the more thoroughly the relevant regularities are understood, the more precisely the individual event can be explained.

point of view [...] constitutes one of the book's most fascinating features" (Meyerson 1911, 100). Cassirer's attempt to integrate the history of science with the philosophy of science was hardly new to French epistemologists (101), but this approach had remained largely foreign to the German philosophical tradition, with the notable exceptions of Friedrich Albert Lange (1873) and Kurd Lasswitz (1890). Only recently had this approach begun to gain renewed prominence in Germany, "under the sustained influence of Hermann Cohen's school." In light of this achievement, the *Erkenntnisproblem* stood as "the most important work produced to date by a disciple of the master of Marburg" (Meyerson 1911, 101). Meyerson praises Cassirer's remarkable erudition, which extends beyond the history of philosophy to encompass a wide range of developments in the history of science. In Meyerson's view, Cassirer's book was both a significant innovation and a model for the historical study of scientific thought, an approach that was far less familiar to philosophers and historians of philosophy at the time than it is today.[18]

Nevertheless, Meyerson could not refrain from pointing out a glaring shortcoming in Cassirer's book: "The author has not, it seems to us, paid sufficient attention to the history of chemistry." Cassirer's analysis could have drawn "great advantage" (102) from the contribution of chemists to the problem of knowledge. In this regard, Meyerson seems surprised that Cassirer did not take into account the four volumes of Hermann Kopp's (1869) history of chemistry which would have considerably enriched the historical background of the *Erkenntnisproblem*.[19] In Meyerson's view, Cassirer's omission of the history of chemistry appeared to be more than a mere oversight. Meyerson suspects that, in order to justify his systematic premises, Cassirer deliberately obscured the history of chemistry because it did not align with his functionalist account of scientific cognition.

Meyerson concedes that Cassirer might object that chemists remained largely isolated from the broader intellectual movement until the second half of the eighteenth century. Boyle is the only exception, being both a great chemist and a renowned physicist, and Cassirer could reply that he had in fact devoted considerable attention to Boyle's work. However, in the second volume of the *Erkenntnisproblem*, in fact, Cassirer discusses Boyle's contribution to epistemology, referring solely to his anti-Aristotelian work *De ipsa natura*, ith having replaced the Renaissance organicist conception of nature redefining nature as a "system of rules" (Cassirer 1907, 312–313) However this interpretation, for Meyerson (Meyerson 1911, 125), entails an unforgivable "mutilation" of Boyle's thought. Upon closer examination, Boyle was not simply the supporter of a lawful conception of nature; he was -"a strict atomist" (122) and a proponent of a "corpuscular" theory of matter (an expression coined by Boyle himself): "His treatises are filled with it, and a large part of his activity was devoted to demonstrating the proposition that all changes can be due only to the displacement of molecules or corpuscles." Because Cassirer overlooks Boyle's corpuscularism, his interpretation appears to Meyerson "incomplete and therefore, in a certain sense, misleading" (124). Boyle was not seeking mere abstract mathematical rules, but the true physical causes and ontological reality behind phenomena.

If Cassirer had considered Boyle from this point, he would recognize the functional *legality* of nature, but also the centrality of the concept of *substance*. It is true that

[18]On this point, see also the lecture Meyerson delivered on January 25, 1911, at the École des Hautes Études Sociales (Meyerson 2011, 195–197).

[19]Hermann Kopp had been Meyerson's professor in Heidelberg between 1878 and 1880. (Bensaude-Vincent and Telkes-Klein 2016, 31).

Cassirer does not resort exclusively to laws, by means of that notorious "subterfuge" of Comte and Mach, of claiming that atomists behave in a purely instrumentalist way, as if atoms exist: "Rather, he seems to arrive at a system related to the logical theories of the Marburg school, in which the particular form of physical hypotheses would be justified through the application of the concept of function" (Meyerson 1911, **). As further evidence of Cassirer's attempt to conceal the history of chemistry, Meyerson (103) notes that Lavoisier the father of quantitive revolution of chemistry is never mentioned in the *Erkenntnisproblem*. This omission, like that of Boyle's corpuscularism, did not seem coincidental. In Meyerson's view, Lavoisier's contribution to chemistry stood in radical conflict with Cassirer's depiction of the history of modern science. Rather than dispensing with the concept of substance in favor of a mathematical understanding of chemical transformation, Lavoisier systematically relied on the notion of the "thing in itself" to challenge the theory of phlogiston (Meyerson 1908, 202–207/222–227). Kant himself, Meyerson (Meyerson 1911, 103) insists, was strongly influenced by Lavoisier, but Cassirer failed to acknowledge it for the sake of his own epistemological position.

The contrast between Cassirer and Meyerson ultimately turns on their different assessments of the relationship between scientific knowledge and "naïve realism." For Cassirer, the development of science involves a progressive departure from the assumptions of ordinary experience. However, in this way in a form of crypto-positivism, that "purely phenomenal reality" or a "world of relations without a support [*un monde de rapports sans support*]." But is entirely foreign to scientific practice. When science destroys an old concept of substance (such as phlogiston), it does not reduce reality to pure mathematical functions; rather, it immediately posits a new objective substance (such as oxygen or atoms) to explain the phenomena. In his view, Cassirer's determination to reconstruct the history of science in accordance with his philosophical commitments leads him to "over-interpret" the historical evidence (121). The consequences are both historiographical and epistemological. On the one hand, Cassirer's reconstruction marginalizes aspects of scientific practice that do not fit his systematic interpretation. On the other, it unwittingly come to share the "positivist" conception of science as a system of laws and functional relations, neglecting the persistent role of substantial entities in scientific explanation (Meyerson 1908, 1–46/17–62).

3.2 The Erkenntnisproblem *Revisited*

One may reasonably assume that, had Meyerson been familiar with *Substanzbegriff und Funktionsbegriff* when he reviewed the *Erkenntnisproblem*, he would not have raised this objection in the same terms. As we have seen, Cassirer not only devoted an entire section to concept formation in chemistry but also accorded it a distinctive "methodological interest" which is precisely the opposite of what Meyerson see to suggest (Cassirer 1910, 292/221). According to Cassirer, chemical concepts show that scientific concept formation need not erase the distinctive characteristics of particular substances (299/224). On the contrary, as chemical concept formation advances, it enhances our ability to distinguish particulars. From Cassirer's point of view, Meyerson, like Rickerts, implicitly assumes that concepts are formed through generic abstraction, that is, by progressively discarding individuals' distinguishing features. The more universal a concept becomes, the further it moves from empirical reality, opening an unbridgeable gulf between the concepts of natural science and the reality they seek to describe. Whereas Rickert draws from this premise the methodological conclusions about the limit of scientific concept formation, Meyerson carries it to its ultimate

metaphysical consequence; like Rickert an 'irrational' residue, that is, a residue that is irrational not in an absolute sense, but only in relation to the intellect's 'discursive' capacity[20]

For Cassirer, Meyerson's philosophical stance did not arise from his interest in the history of chemistry; rather, he could probably retorded the accusation that historiography of chemistry was shaped by his misleading conception of scientific concept formation. If Cassirer had neglected Boyle and Lavooiser, Meyerson was just as culpable to have downplayed the role of Richter and Dalton. Nevertheless, Cassirer took Meyerson's criticism seriously. There is evidence that Cassirer read Meyerson's review of the *Erkenntnisproblem* shortly after its publication. In late February 1911, he sent Meyerson a brief letter of reply:

> I have just read your review of the *Erkenntnisproblem* in the library, and I would like to express to you my sincere thanks. Of all the reviews my book has received—most of which you are familiar with—yours is, in every respect, the most stimulating and helpful. For you entirely grasp the whole work from the fundamental systematic perspective in which it was written, and thus your comments consistently address those points that were, also for me, the most important and truly decisive ones. Of course, you already know that I cannot agree with you on everything, but your critique contains such a sharp and concise formulation of the contrast between us that it already contributes significantly to a mutual understanding. (Cassirer to Meyerson, Feb. 21, 1911; ECN, Vol. 18 (DVD), Doc. 161)

Unfortunately, Meyerson's review appeared too late to be considered for the revision of the first volume of the *Erkenntnisproblem*, whose second edition, at the time Cassirer wrote this letter, had already been published. Nevertheless, Cassirer assured Meyerson that he would pay close attention to his comments in revising the second volume. Before proceeding with its publication, Cassirer also visited Meyerson in Paris at the end of March 1911, thereby taking the opportunity to discuss the more delicate issues of the *Erkenntnisproblem* directly with him..[21]

The second edition of the second volume of the *Erkenntnisproblem* was published in the autumn of 1911. The extensive chapter devoted to the development "from Newton to Kant" includes an entirely new section on the "problem of matter" in modern chemistry (Cassirer 1911, 2:428–441) addressing the shortcomings highlighted by Meyerson. A letter to Meyerson from October 1911 confirms that Cassirer introduced them precisely in response to his review: "Today you are receiving the second volume of the *Erkenntnisproblem* in its second edition. I would be very pleased if you could occasionally find the time to look through it and share with me your opinion. You will notice for yourself that I have made extensive use of your review, and that it has been extremely valuable to me: for example, I have now added a new section on the development of chemistry, and I have also taken into account some details to which you drew my attention." (Cassirer to Meyerson, Oct. 25, 1911; ECN, Vol. 18 (DVD), Doc. 178).

Cassirer opens the chapter by reiterating that, contrary to modern physics, chemistry dominated the "stage of the medieval view of nature," in which it is always

[20] It seems plausible that Cassirer, after the meeting with Meyerson, decided to add a brief comment on Meyerson's philosophy to one of the final drafts of chapter IX. Meyerson did inded. lthough Meyerson is named only in a footnote, Cassirer is clearly referring to his recent revival of Eleatic metaphysics.

[21] See Cassirer to Meyerson, Mar. 30, 1911; ECN, Vol. 18 (DVD), Doc. 164. Meyerson had also sought to arrange a meeting between Cassirer and Gaston Milhaud, who however was forced to withdraw at the last minute because of family circumstances (EMLF, 644).

certain characteristics of bodies and groups of bodies that, in this conceptual language, are reinterpreted as thing-like entities. However, now the victory of the mechanical conception of nature gives the concept of 'element' an entirely new meaning. Boyle's "corpuscular philosophy" recast the "occult qualities" in terms of "distinctly cognizable properties, such as magnitude, shape, and motion" (Cassirer 1911, 2:430). As a result of this shift, the internal constitution of matter was no longer conceived as an "unfathomable ground" (2:426) inaccessible to human cognition, but rather as an object that could be measured and interpreted in mathematical terms. In this way, Cassirer while acknowledging Meyerson's objection, he could integrate it into his own conceptual framework. In replacing old substantial forms with the system of rules governing the motion of corpuscle, Boyle replace the the old metaphysical substances with the laws of motion: "The *material* concept of nature is thus subsumed (*aufgehoben*) under the *formal* one, nature as a thing under nature as the totality of rules"Der materiale Begriff der Natur ist damit | in den for|malen, die Natur als Sache in die Natur als Inbegriff der Regeln aufgehoben (**). As Meyerson, Cassirer emphasize that Boyle's approach played a little role in the early development of chemistry itself. John Freind (1709), for example, attempted to derive the whole of chemistry by the Newtonian method, seeking to introduce the concept of attraction into it and to make it fruitful for explanation (Cassirer 1911, **). However, these attempts to establish a chemical dynamics remained isolated, as empirical knowledge was not yet sufficiently advanced.

Eighteenth-century chemistry continued to be dominated by the theories of the 'phlogistonists,' who retained a qualitative conception of natural processes. It is through Lavoisier's work that the "scientific concept of chemistry itself is transformed." If *Substanzbegriff und Funktionsbegriff* Lavoisier is mentioned only in passing, in the second edition of the *Erkenntnisproblem*, however, he takes center stage. The decisive innovation lay not in the discovery of new facts, since phlogiston theory had already recognized that combustion could involve an increase in weight, but in the new logical interpretation of those facts. Whereas the phlogistonists regarded quantitative measurements as superficial, Lavoisier made the quantitative relations between the initial and final states of chemical transformations a fundamental theoretical criterion for determining what counts as a 'substance.' Once again, whereas Meyerson regarded Lavoisier as exemplifying the centrality of the substance concept, Cassirer could incorporate Lavoisier into his broader account of the transformation of that very concept through the conversion of chemical experience into exact quantitative determinations. Admittedly, Lavoisier achieved this result by adopting a strictly empiricist approach. Nevertheless, as Cassirer argues, "Lavoisier's achievement implicitly represents one of the most important steps toward the recognition of the logical principles of empirical knowledge itself: for here the concept of substance passes definitively from its ontological into its modern epistemological-critical meaning, being conceived as a pure correlative concept to the idea of quantitative conservation."Dennoch bedeutet Lavoisiers Leistung implizit einen der wichtigsten Schritte in der Richtung auf die Erkenntnis der logischen Grundsätze der Erfahrungserkenntnis selbst: Denn der Begriff der Substanz geht hier endgültig von seiner ontologischen in seine moderne erkenntniskritische Bedeutung über, indem er als reiner Korrelatbegriff zum Gedanken der quantitativen Erhaltung gefaßt wird (**)

The Meyerson–Cassirer 'battle over the substance-concept' was fought with historical weapons, but the battlefield was ultimately a purely systematic one. Cassirer was ready to accommodate Meyerson's objections on historical grounds, though he entirely rejected

their epistemological implications. Meyerson contends that scientists (such as Boyle with his rigid corpuscular atomism and Lavoisier with his conservation of matter) continuously appeal to something that remains constant beyond the apparent variety of the phenomena. By contrast, Cassirer could easily reinterpret the same episodes as confirming his narrative of the progressive reduction of the substance-concept to the function-concept (Boyle's atom moves according to mathematical laws, and Lavoisier) Regardless of this divergence, in November 1911, Meyerson informed Cassirer that he found the second edition of the *Erkenntnisproblem* "considerably improved" (Meyerson to Cassirer, Nov. 15, 1911; ECN, Vol. 18 (DVD), Doc. 180). Meyerson most likely appreciated Cassirer's effort to engage with sources he had not considered in the first edition, even though Cassirer interpreted them within a narrative diametrically opposed to his own. If the Meyerson–Cassirer debate teaches us anything, it is perhaps the limitations of such *grand récit* that are either too narrow to accommodate the complexities of historical detail or, as in this case, so broad that virtually any historical evidence can be made to fit them.

4 Conclusion

After 1911, the correspondence between Cassirer and Meyerson became more sporadic. However, in December 1912, just after the publication of the second edition of *Identité et réalité* (Meyerson 1912), in which he introduced multiple references to Cassirer's works, Meyerson sent him two issues of the Bulletin de la *Société française de philosophie* devoted to the discussion of Perrin's study on Brownian motion. The brief letter accompanying the issues reads as follows:

> At the same time, allow me to send you two issues of the Bulletin de la *Société française de philosophie*, which you probably do not see regularly. They contain Perrin's lecture on the new measurements of atoms, as well as the ensuing discussion. I also took part in the discussion and, of course, I attempted to turn the situation to my own advantage. Although you will probably disagree with almost everything, you might perhaps be interested in certain aspects of the whole subject. Should you write anything about it (for which I would be very grateful), I kindly ask you to send me a copy. (Meyerson to Cassirer, Dec. 1, 1912; ECN, Vol. 18 (DVD), Doc. 195)

Up to Perrin's (1910) work, the amounts of that element entering different molecules occur in integer multiples of its relative atomic weight. These relations leave the scale individual atomic masses undetermined Perrin's (1910, 268), since they do not establish Avogadro constant N_A—ratio between an amount of substance and the number of particles that it contains. Perrin showed that Brownian motion, electrical measurements, radioactivity, and other phenomena yield mutually compatible determinations of N_A. Perrin takes the fact that this measurements "converge precisely on the same number" (281) this as evidence for the actual existence of the atom each endowed with absolute weight. Thusc the Brownian motion of small suspended particles demonstrates directly to our senses the presence of a molecular thermic motion.

For Meyerson, Perrin's result showed that, despite the prestige of positivist ideas, science could not avoid reintroducing atoms as microscopic 'things,' beyond the functional relations among measurable quantities. Cassirer's response to Meyerson's letter reveals a cordial yet detached tone: "Thank you very much for sending me the report on the discussion at the *Société française de philosophie*; I read it with great interest." (**) However, he did not add further comments. Indeed, after the publication of *Substanzbegriff und Funktionsbegriff*, Cassirer moved away from 'philosophy of science'

until he wrote his relativity book a decade later (Cassirer 1921). Here, Cassirer, only add sporadic to the role of atoms in chemistry, However, his stance seems to have been remained unchanged. The importance of chemistry in the history of the concept that constancy of atoms as thing, becomes the constancy in the constancy of a numerical value: "fundamentally, such a realization, such a strictly quantitative meaning of the concept of the atom, was only reached in the beginnings of modern chemistry in the law of multiple proportion" (**; tr. 1923, **). As we have seen, according to Cassirer, with the further help of Avogadro's Law, Dulong and Petit's Law, and Mitscherlich's Law, chemistry could remove the ambiguity in Dalton's law by progressively and univocally determining the *relative* weights of atoms with respect to other atoms. Perrin's result could be seen as part of this process of further disambiguation. By fixing the exact scaling factor, Avogadro's constant, it becomes possible to univocally determine the *absolute* weight of an individual atom in grams. In Cassirer's view the 'reality' of the chemical atom is nothing but this convergence of progressively more stringent numerical determinations: "the atom shows itself thereby to be, not an absolute minimum of being, but a relative minimum of measure" (Cassirer 1921, 18; tr. 1923, 360).

Altug, share some anti-realist Cassirer do deny that atoms exitst, bit that atoms exit, but nature atomc-oncet Cassirer points out that the absoluteness of this 'minimum' always provisional to a certain stage of the development of science and cannot exist. Durch die Erkenntnis der radioaktiven Verschiebungs|sätze discovery that 'isotopes' of the same element have different masses. The atomic weith gives average mass per atom when M is the molar mass of a mixture of differnt isotes . Their Individual identity is not univocal determ atomic mass, Atomic weight could no longer be retained as the true ordering principle of the natural system, it proved "unexplained function of the true "atomic number."" (Cassirer 1936b). This true atomic number based on Bohr's atomic model, in a neutral atom, the number of electrons equals the number of protons (Cassirer 1929, 516; tr. 1957, 442–43). could the show that. Each element is again a position in a sequence, but now according to a deeper sequence principle *Reihenprinzip*: "This principle itself is never simply 'given' [*gegeben*] but represents a perpetual problem [*aufgegeben*], and one of the essential achievements of all natural science is to be sought in a steady advance toward a more complete solution of this problem" (Cassirer 1929, 518; tr. 1957, 444).

Bohr's atomic model made it possible, for the first time, to gain a firm foothold, so to speak, in atomic physics, derivations of Balmer's equation, the Rydberg constant, Ritz's combination principle, as well the periodic system. Here any doupt on the exitecen of atoms had t removed. However, for Cassirer, the distinction between 'phenomenalism' and 'realism' was once again the wrong fault line: "We are concerned not so much with the existence of things as with the objective validity of relations; and all our knowledge of atoms can be led back to, and depends on this validity" (Cassirer 1936a, 178; tr. 1956, 143). As one migth expect from a Kantian, the empirical reality of atomic *entities* resolved completely objective validity of the atomic *theory*. However, for Cassirer this 'objective validity' is never achieved as ultimate, but it is always relative to a certain stage of the history of science. The succession of physical theories attribute to the atom a progressively more determinate form: "But this 'concrescence' [*Konkreszenz*] of determining factors stands in an inverse relation to what we commonly designate as visualizable concretion [*Konkretion*]" (Cassirer 1936a, 178; tr. 1956, 143). However, despite the loss of intuitive, pictorial models, physical 'objectivity' is thereby increased rather than diminished, for "true physical objectivity must never be sought in the

direction of 'naive' things, but rather in the direction of the critical concept of law [*Gesetzesbegriff*]" (Cassirer 1936a, 178; tr. 1956, 144; translation modified).

If Cassirer borrw some anti-atomistic arugem, his concern is not the atocm realim atoms, but the fact that atom-concet is not a sustacen-concet. The entire history of the atom-concept represent for Cassirer a confirmation of this view. Here one observes indeed "*consolidation* of the atom-concept", but not an "*hypostatization*, its immediate reification" (Cassirer 1936a, 171; tr. 1956, 144; translation modified). In the sections dedicated to the history of the chemical atom, Cassirer had now clearly integrated some of Meyerson's suggestions, pointing out the role of Boyle's corpuscularism is part of the hitory of quantitative chemistry fro the outset. However, that the Dalton's atomism chemical atom aquate aemprical realti of multy emrical alws, conserfavtion of matter to multimep pproptions:

> [T]he atom is here defined strictly substantially; but all statements concerning its nature (*Was*) ultimately merge into statements concerning quantity-definite numerical relations (*Wieviel*). All the basic concepts and theories of chemistry, the concepts of valence and types, the concept of the radical as well as the theory of compound radicals all point in this direction. The conceptual language of atomism in conjunction with the general symbolic language of mathematics now permits that simple and magnificent development in the language of chemistry that is evident in the progress from simple to structural formulas. Stereochemistry, as established by van't Hoff, permitted the elucidation, through a complex construction of formulas, of an entirely new group of facts, the phenomena of stereoisomerism; it threw new light on the distribution of atoms in space. Thus the visual picture of the atom is modified and refined, but all these changes serve only the one purpose of circumscribing and representing an increasingly large domain of functional relations. (Cassirer 1936a, 183–84; tr. 1956, 147)

The discovery of the electron turned the chemical atom of Dalton into a composite object (Thomson's 'plum pudding' model) and the discovery of the nucleus fixed into a spatially structured object (Rutherford's nuclear atom and Bohr's planetary model). From this point of view, as Cassirer put it, the history of the atom-concept appears a "kaleidoscopic" variety, a strange "phantasmagoria" of disconnected portraits. However, the history has good reason to see our sequence as correcting errors of former members while preserving their successes "in a single direction" (Cassirer 1936a, 187; tr. 1956, 150). It is this accumulation of historically stable properties that we call the 'reality' of the atom: "Thus we need not find an answer to the question of what the atom actually and ultimately is, as long as we are sure that the various forms of determination given us by progressing experimental and theoretical investigation can dovetail and mutually supplement one another" (Cassirer 1936a, 187; tr. 1956, 150). The 'atom,' therefore, is not merely a subjective device for achieving economy of thought, as the positivists claim; but neither does it possess a thing-like existence, as the realist It is these historically stable properties that endure through subsequent theory changes. This accumulated stock of enduring properties can be collected into what we shall call the structure of electron theory.

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